

Randomized and low-rank approximation

Constant-loss nuclear-error transfer without matrix ordering

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Historical assessment of the original question: challenging because a dimension-independent loss must survive removal of matrix ordering; community impact is transferring nuclear-error guarantees between matrix functions.

Topic: Low-rank approximation of matrix functions.

Resolution - 2026-09-11

Author: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA.

Affirmative resolution: the universal constant $C = 11$ works. [Theorem 1 of the complete proof](#) covers every matrix size, allowed rank, real PSD pair, nonnegative continuous operator-monotone function and $\varepsilon \geq 0$ in the original statement. It retains every permitted choice of ordered eigenvectors, including ties, selected zero eigenvalues, $f(0) > 0$ and zero optimal tail. No ordering or commutation assumption is added. No case of the canonical nuclear-norm question remains open; optimality of 11 and other norms are not claimed.

[Proof PDF](#) · [Standalone TeX](#) · [Independent Codex-agent review: PASS](#) · [Authorship, source identities and submission record](#).

The new estimate is the compression bound in **Lemma 2**; Sections 3-5 give the spectral replacement, constant 11 and positive-integral reductions. The work used substantial ChatGPT/Codex assistance. The separate agent review and coordinating-agent source audit are documented; external human peer review and formal certification are not asserted. The original statement, ID and historical ratings are retained below.

Earlier partial result - 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review: PASS for the limited result below.**

The sharp nuclear relative-excess factor is two when $A, B = \hat{A}_k$ and the actual selected rank- k projector have a simultaneous orthonormal eigenbasis. The finite-Schatten extension is also proved. Diagonal operator-monotone power examples show that any constant solving the full question must satisfy $C \geq 2$.

Historical scope: At the time of this partial result, existence of a finite universal constant for arbitrary noncommuting PSD pairs remained open. The full nuclear-norm question is answered above with $C = 11$. Commutation of A and $\hat{A}$ alone does not cover every truncation inside a repeated eigenspace. The separate scalar-concave nuclear counterexamples use functions outside the required operator-monotone class.

Primary reference: [complete authored PDF](#), [standalone TeX](#), **Theorem 1.1 and Section 3**. [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Does a universal constant $C \geq 1$ exist with the following property? For every $n \geq 2$, $1 \leq k < n$, real symmetric positive semidefinite matrices $A, \hat{A} \in \mathbb{R}^{n \times n}$, $\varepsilon \geq 0$, and continuous operator-monotone function $f : [0, \infty) \rightarrow [0, \infty)$,

$$\|A - \hat{A}_k\|_* \leq (1 + \varepsilon)\|A - A_k\|_*$$

implies

$$\|f(A) - f(\hat{A})_k\|_* \leq (1 + C\varepsilon)\|f(A) - f(A)_k\|_*$$

Here $\|M\|_* = \sum_i \sigma_i(M)$ is the nuclear norm. Operator monotonicity means $f(H) \geq f(G)$ for every size and every real symmetric $H \geq G \geq 0$; functions of matrices are defined spectrally. No ordering between A and $\hat{A}$ is assumed.

For an ordered orthonormal eigendecomposition $X = \sum_{i=1}^n \lambda_i q_i q_i^T$ of a PSD matrix, set

$$X_k = \sum_{i=1}^k \lambda_i q_i q_i^T, \quad f(X)_k = \sum_{i=1}^k f(\lambda_i) q_i q_i^T.$$

The assertion must hold for every choice of ordered eigenvectors, using the same choice in both truncations. C must be independent of all inputs, including f .

This is the nuclear-norm instance of the authors' explicit question about a fixed loss in approximation accuracy. It asks whether the ordering condition in existing transfer results can be removed at a controlled price.

References

1. D. Persson, R. A. Meyer, and C. Musco, [Algorithm-agnostic low-rank approximation of operator monotone matrix functions](#), *SIAM Journal on Matrix Analysis and Applications* 46 (2025), 1–21. §1.2, paragraph immediately preceding Table 1; §5, Example 5.3 and closing paragraph; Theorem 2.4 is the ordered predecessor. Equation (2) specifies truncation.
2. D. Persson, T. Chen, and C. Musco, [Randomized block-Krylov subspace methods for low-rank approximation of matrix functions](#), *Linear Algebra and its Applications* 741 (2026), 32–65, abstract and algorithmic scope.

Historical status check - 2026-09-08

Checked the source’s latest listed arXiv v2 (2024-07-04), its 2025 journal record, and the exact counterexample and constant-loss discussion. Searches combining the title, “constant”, “counterexamples”, “nuclear”, “funNyström”, and 2026 found no general resolution. Example 5.3 rules out $C = 1$, while the source expressly leaves a larger fixed constant possible. The 2026 Krylov work addresses specific algorithms, not arbitrary PSD approximation pairs. This entry is explicitly a specialization of the source’s broader question; the source does not number it as a separate conjecture. No later explicit reaffirmation was located.

Historical audit - 2026-09-10

Rechecked [Theorem 2.4 and Example 5.3](#): ordered pairs satisfy $C = 1$, while unordered pairs can violate that constant. At that audit, a larger universal constant remained open. Constant-loss nuclear-transfer searches found no resolution; the partial tag refers only to the ordered subclass.